

TUROCHAK, Russian Federation

WMO: 360610

Lat: 52.250N

Lon: 87.127E

Elev: 327

StdP: 97.46

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.9	-31.5	-37.3	0.1	-34.8	-33.8	0.2	-31.4	5.9	-4.7	4.4	-10.2	0.0	210	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	30.7	19.3	28.9	18.5	27.1	17.7	20.7	28.0	19.7	26.5	18.9	25.0	1.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.8	22.8	17.5	13.0	21.9	16.6	12.3	21.3	61.3	28.2	58.2	26.5	55.1	25.1	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.3	3.6	-37.7	34.6	3.5	1.8	-40.3	35.8	-42.3	36.9	-44.3	37.9	-46.8	39.2		
(5)			-37.7	22.5	3.1	1.3	-39.9	23.5	-41.8	24.3	-43.5	25.0	-45.8	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-17.1	-12.7	-4.1	4.5	11.5	17.4	19.4	17.2	11.0	3.7	-6.4	-14.3		
	DBStd	13.69	6.92	6.62	6.14	5.27	4.59	3.54	2.87	3.33	4.09	4.70	7.04	6.88		
	HDD10.0	3608	840	636	436	175	38	1	0	1	37	200	492	753		
	HDD18.3	5860	1098	869	694	416	215	57	22	61	222	453	742	1012		
	CDD10.0	905	0	0	0	9	85	223	291	224	67	6	0	0		
	CDD18.3	115	0	0	0	0	4	29	54	26	2	0	0	0		
	CDH23.3	1539	0	0	0	11	134	445	566	338	45	1	0	0		
	CDH26.7	441	0	0	0	1	31	126	178	97	8	0	0	0		
	WSAvg	1.1	0.5	0.6	1.1	1.5	1.6	1.3	1.2	1.2	1.2	1.2	1.0	0.7		
	PrecAvg	821	36	32	37	68	82	88	96	92	80	68	71	65		
(15) Precipitation	PrecMax	981	65	56	78	151	168	148	178	199	161	146	139	179		
	PrecMin	593	11	8	8	23	43	37	53	46	29	16	30	16		
	PrecStd	94	14	12	17	39	32	32	31	39	37	31	32	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.4	4.0	13.3	24.7	30.0	32.1	33.1	33.1	27.8	20.4	10.0	0.9		
		MCWB	-0.5	0.5	6.2	12.2	16.6	19.0	20.8	20.0	17.1	11.9	5.9	0.3		
	2%	DB	-2.6	1.3	10.3	20.3	27.1	29.9	30.7	29.4	24.2	16.7	6.3	-1.2		
		MCWB	-3.3	-1.0	4.4	10.4	14.7	18.4	20.0	19.4	15.5	10.0	3.5	-1.8		
	5%	DB	-4.8	-0.8	7.5	17.0	24.6	28.0	28.8	27.0	21.3	13.5	3.6	-2.7		
		MCWB	-5.4	-2.4	2.9	9.1	13.7	17.6	19.4	18.4	14.1	8.7	2.1	-3.2		
	10%	DB	-6.9	-2.9	4.7	13.9	21.7	26.0	26.8	24.9	18.8	10.6	1.6	-4.5		
		MCWB	-7.5	-4.3	1.3	7.5	12.9	17.1	18.6	17.6	12.9	7.3	0.6	-5.0		
	0.4%	WB	-0.1	1.0	6.8	13.1	17.8	20.8	22.4	21.6	17.8	12.6	6.4	0.4		
		MCDB	0.4	2.9	12.7	22.3	27.7	28.8	29.5	30.1	26.1	19.0	9.4	0.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-3.2	-0.7	4.7	11.3	16.2	19.6	21.1	20.1	16.0	10.8	3.8	-1.7		
		MCDB	-2.7	1.0	9.3	18.4	24.0	27.1	28.3	27.9	22.8	15.4	5.9	-1.2		
	5%	WB	-5.3	-2.4	3.2	9.8	14.8	18.7	20.2	19.0	14.7	9.1	2.1	-3.2		
		MCDB	-4.7	-0.8	7.0	15.6	21.7	25.5	26.5	25.5	20.3	12.9	3.4	-2.7		
	10%	WB	-7.4	-4.3	1.6	8.1	13.6	17.9	19.3	18.0	13.4	7.5	0.7	-5.0		
		MCDB	-6.9	-3.0	4.3	13.6	19.7	24.0	25.2	23.5	17.9	10.5	1.5	-4.5		
	5% DB	MDBR	11.6	14.9	14.7	12.7	13.5	12.8	11.6	11.9	12.3	10.0	8.6	9.7		
		MCDBR	9.4	15.2	17.8	19.4	20.2	17.7	16.2	17.1	18.1	16.7	9.4	7.8		
		MCWBR	8.6	12.6	11.9	10.0	8.6	7.2	6.4	7.5	9.1	9.7	6.6	7.3		
		MCWBR	8.8	14.4	15.9	17.0	15.6	14.5	13.5	15.1	16.3	15.3	8.8	7.4		
(40) Clear-Sky Solar Irradiance	taub		0.294	0.333	0.385	0.434	0.419	0.411	0.426	0.409	0.373	0.367	0.334	0.302		
		taud	2.039	1.975	1.933	2.034	2.185	2.302	2.282	2.299	2.339	2.220	2.075	1.989		
	Ebn at Noon		659	740	777	794	832	843	822	815	803	709	599	558		
		Edh at Noon	79	115	148	152	137	123	124	116	99	90	77	71		
(44) All-Sky Solar Radiation	RadAvg		1.19	2.25	3.60	4.49	5.26	6.01	5.61	4.72	3.46	1.89	1.01	0.81		
	RadStd		0.13	0.17	0.35	0.36	0.54	0.52	0.33	0.34	0.30	0.21	0.13	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page